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AI-Integrated CCTV System for Hazard Detection with Tamper-Evident Logging in Malaysian Nursery and Child Care

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Links:

i. GitHub Repository: https://github.com/ttwceline/MAL3018_CTTWCCTV.git
ii. PebblePad: https://v3.pebblepad.co.uk/spa/#/public/mtw79Zy757466Mmh35bs58bp9h
iii. Google Drive: https://drive.google.com/drive/folders/1rnruvb1utsjvWOiUvC1Y7wEa9cGNLc73?usp=sharing

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1.0 Introduction

1.1 Background

In the environment of a nursery, the people placed there are those who do not have the ability to report any mistreatment towards them. Babies, infants and toddlers are not capable to express neglect and discomfort. Moreover, nobody would bother nor care if a crying baby is not attended to unless there are proactive monitoring. In this setting, Conventional CCTVs holds multiple weaknesses:

1. **Reactive rather than Proactive**. Usually, a footage is only reviewed after a complaint. It is not realistic for a caretaker to be giving continuous attention as they are occupied with multiple children at the same time doing variety of chores; feeding, changing and assisting.
2. **Footage is not verifiable**. A footage can anytime be trimmed, replaced and even deleted. Without any cryptographic evidence, the parents, owners and operators do not have a way to prove the integrity of a clip. It weakens the trust among all the parties involved. This includes the caretakers that would like to prove themselves that they were properly doing their job in any cases of accusations.

Childcare centres catered for children below the age of 4 are regulated under the Child Care Centre Act 1984 (Act 308) and the Child Care Centre Regulations 2012 that is administered by Jabatan Kebajikan Malaysia (JKM). There are public statements by the Women, Family and Community Development (KPWKM) former Minister, Datuk Seri Rohani Abdul Karim, stating that installing CCTVs in daycare centres and kindergartens are the best option to ensure safety of the children due to mismanagement and abuse cases that were on the rise. After the statements, in August 2023, the current Minister, Datuk Seri Nancy Shukri stated that there were some nursery premises that have not installed CCTVs as there was no law enforcement so far. The same statement indicates that there were about 3,767 cases of children that are in need of care and protection nationwide in the first six months of 2023 (Bernama, 2023). Due to the events, The Social Welfare Department took actions of making it mandatory for CCTV installations in legally registered nurseries as an effort to improve operations of childcare centres (Mohammad, 2023).

The primary users of the system are the nursery principal that will receive a first alert and must take action upon receiving an alert. Besides, there is the nursery owner who gets notified when the alerts get unanswered and reviews any escalation report. Caretakers on the

other hand are indirectly involved as the nursery principal are assigned to direct them to attend to the children in need. As for the parents, they are indirect beneficiaries who may gain verified records in any occasions of cases. The system does not identify any child or staff by their name or personal records.

1.2 Problem Statement

1. Conventional CCTV systems in Malaysian nurseries are reactive rather than proactive. Whenever there is a baby who has been crying for some time, it may be unnoticed unless there is someone who is aware of every single child. In any cases of neglect, footages are only reviewed after complaints.
2. Existing commercial baby monitors are just built for house usage and it depends on ven-dors clouds. It does not provide any independently verifiable records. Besides, cur-rent products available in market only detects the crying of babies without tracking the dura-tion.
3. Footage and event records are unverifiable. The footages and logs have potential of being edited, deleted and replaced without any proof of tampering. This weakens the trust be-tween parents, caretakers and regulators.

This project goal is to provide **an open-source system** that **can detect the movements of the children** within a nursery with an additional feature of **detecting crying** infants and **rec-ognising the crying episode** if it has **lasted five minutes and above**. It will then alert the nursery principal, and, if the alerts are ignored, the case is escalated to the owner of the nursery through a defined and recorded chain of responsibility. Lastly, the **footages and clips** are **en-crypted** to keep **tamper-evident records** to ensure integrity and enable verification of records.

2.0 Objective

I. **To develop an AI-Integrated close-circuit television system that detects hazard like falling, abuse, violence.**

- To immediately alert whenever a child is at a potentially dangerous position.
- To avoid any accidents that could be fatal.

II. **To implement infant crying detection via camera audio with additional duration tracking that alerts person-in-charge after a prolonged period of time.**

- Prolonged crying may cause raised levels of cortisol in an infant.
- The hike in cortisol heightens anxiety, disrupt sleep cycles and alters brain's neurotransmitter system that causes structural and functional changes in the brain of the infant similarly to depressed individuals (AskDrSears, n.d.; Abe, 2026).
- Based on a statement by Abe, crying that lasts beyond 5 to 10 minutes are recommended to be checked in for a closer look by most paediatrics. (Abe, 2026)

Figure 1 shows the general guidelines for baby crying duration based on age.

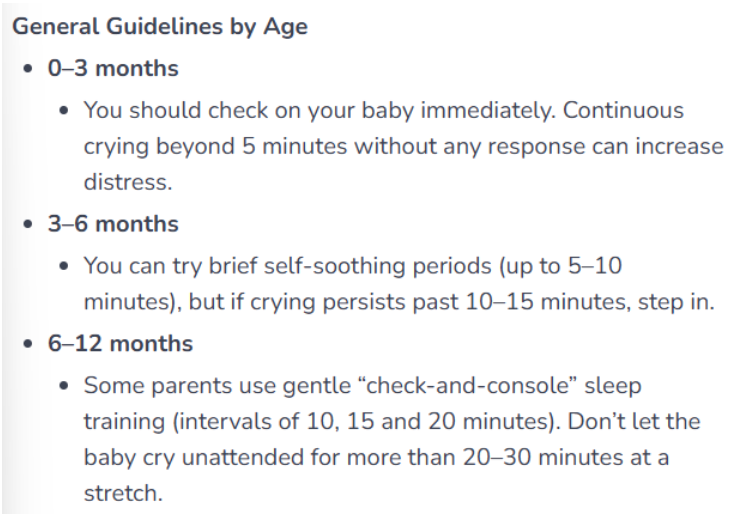


Figure 1: How long is too long to let a baby cry? (Abe, 2026)

III. **To implement a tamper-evident log using SHA-256 hashing chaining, AES-256 (GCM) encryption and Ed25519 signatures that records the detections, responses and alerts.**

- Ensuring there's no tampering of footages as the evidences need to be proved with integrity.

3.0 Scope of the Project

In Scope

1. A single-camera (extendable to two or three) near real-time pipeline on a local machine, simulating a nursery room.
2. Crying-episode tracking based on duration and an escalation workflow. The principal will receive a first alert at the moment the crying duration hits 5 minutes. Repeated alerts of 2-minutes response window if unanswered. The moment the alert goes off for more than 3 times, the owner of the nursery receives a notification via Telegram.
3. Prototype notifications are through a dashboard interface for the principal.
4. Prototype notifications are through a messaging channel, Telegram for the nursery owner.
5. Local, lightweight tamper-evident logging using SHA-256 cryptographic hash chain, AES-256 encryption algorithm and Ed25519 signatures for stored footage to detect edit, delete, insert and reorder tampering.
6. Acknowledgement and false-alarm responses from the principal, and a signed escalation report for the owner.
7. Real-time footage detection on an interface showing timestamp, incident alert, footage captures and human-in-the-loop alert verification.

Out of Scope

1. Medical or diagnostic claims of any kind, including cry-cause classification (hunger, pain, etc.) as a clinical tool, vital-sign monitoring, or any use as a safe-sleep or SIDS monitor. The system does not work as a medical assistant device and does not replace supervision.
2. Face recognition and biometric identification of children or staff.
3. Mobile application interface.
4. Fully functioning UI/UX for the entire dashboard, only focuses on main objectives.
5. Large-scale deployment.
6. Verbal, social, psychological abuse are types of abuse that will not be tracked as it cannot be detectable by camera.

4.0 Literature Review

In this section, it evaluates the context of the project and compares the proposed system to existing systems that are currently in the market.

4.1 Product Review

A) CuboAI Smart Baby Monitor

The CuboAI Smart Baby Monitor is a bird-shaped, AI-centric smart monitored that is specifically used to for prevention of infant sleep accidents. It focuses on paediatric sleep health. The camera operates via a 2.5k QHD camera that is equipped with an infrared night vision and climate sensors. The system has additional features of logging sleep, wake-sleep timelines and milestones for a digital baby diary. CuboAI has its constraint on a premium hardware price tag. The product uses AES-256-bit symmetric encryption for its security standard (CuboAI, 2024).

B) Nanit Smart Baby Video Monitor

The Nanit Pro Smart Baby Video Monitor is a infant monitoring system that functions through clinical data tracking, sleep science and a structured nursery hardware. The monitor is positioned overhead through a dedicated stand. The camera is placed in a birds-eye setting of the entire crib and has integrated two-way audio, ambient sound machines and a soft nightlight. This system continuously monitors the breathing of the infant and sounds alarms when there are irregularities.

C) Xiaomi Smart Camera C500 Pro

The Xiaomi Smart Camera C500 Pro has a 5MP sensor that captures 3K quality video containing HDR support. Besides, it has low light mode that aids in full-colour recording in environments that is not bright. The features of the camera is it supports human detection, pet tracking, baby crying alerts and loud sound detection. The alerts are sent through a dedicated Xiaomi Home Application. The storage for the clips is a MicroSD card (8GB to 256GB), NAS network storage and its dedicated encrypted cloud backups.

D) Proposed System – Little Guard

In the proposed system, it focuses on a system that is applicable to varied cameras, webcams and IP Camera as long as the software of the hardware is compatible with the proposed system. Within the system, the features are **hazard detection** where it can detect

falls, acts of abuse, crying and violence detection. The uniqueness of the system is that it is able to **detect the duration of the crying** of an infant/child. The duration of the cries is monitored and once it reaches the **timing threshold**, an **alert** will be sent to the web dashboard and through a Telegram alert. Whenever the alert is ignored for more than **3 times**, a report alert will be sent to a higher up of the institution. It is a system that focuses on **nurseries and even child-care like babysitting**. Additionally, the system encrypts the footages with **SHA-256 hashing chaining** and **AES-256 (GCM) encryption. Ed25519 signature** is to be implement for it designated hardware.

4.2 Product Comparison

Feature/ Criteria	CuboAI Smart Baby Monitor	Nanit Smart Baby Video Monitor	Xiaomi Smart Camera C500 Pro	Proposed System – Little Guard
Dashboard Interface	Yes	Yes	Yes	Yes
Focuses on Nurseries	No – focused on homes.	No – focused on homes.	No – focused on homes.	Yes – multipurpose.
Type of Detection	• Covered faces • Rollovers • Crying • Cough	• Crying • Motion • Cough • Baby standing	• Crying	• Crying, with duration-based rule. • Physical Abuse • Motion • Falls
Mobile Application	Yes	No	No	No specific application but there is alert system that directly goes to Telegram.
Cost Per Month	$359	$289.99	$45 - $65	N/A

Utilization in Malaysia	Yes	Yes	Yes	Yes – will be focused on Nursery sectors in Malaysia.
Real-time Alerting	Yes	Yes	Yes	Yes
Human-in-the-Loop	No	No	No	Yes
Tamper-free Footage Integrity Verification	Yes – **AES-256-bit** symmetric key encryption.	Yes – **AES-256-bit** symmetric key encryption.	• Yes – **AES-128-bit** encryption. • Each **built-in MJA1 security chip** is provisioned with an exclusive private key and unique certificate.	• Yes – **SHA-256 hash chain** with **Ed25519 signatures** to write a tamper-proof audit log. (Chain of Custody) • Encryption with **AES-256** for storage of recorded footage.

Table 1: System Comparison Table

4.3 Research Gap

Usually, the implementation of the detections and tamper-evident logging are designed as individual problems. Furthermore, the nursey setting has a mixture of regulatory, small budget and sensitive subjects that has to be addressed. This project is a contribution to an open-source prototype that detects hazardous events, and escalates any unanswered crying through a defined and recorded chain of responsibility. Records are produced for the sake of integrity in cases where third parties (e.g. Parents and Guardians) want to verify due to lack of trust towards the operator.

5.0 Methodology

5.1 Development Approach

This project follows the Agile Methodology. It follows an iterative, sprint-based approach. Every module is built and tested in a sprint before integration. In every sprint, there will be a follow-up with the assigned Supervisor and any required improvements will be made after every sprint is discussed on (Gurnov, 2026).

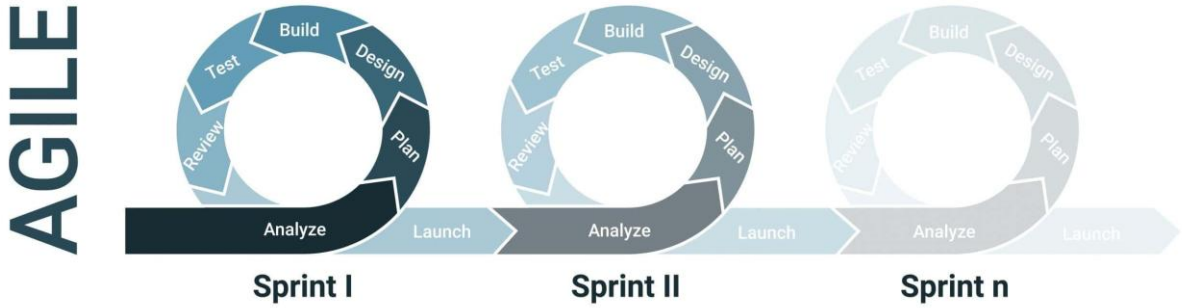


Figure 2: Agile Methodology: What It Means (Goddard, 2021)

The Agile Methodology approach is suitable for this project as it is a safer route. Accuracy of pose-based violence recogniser cannot be known in advance since the violence poses can be confused with rough play. Besides, the frame rate and latency of the camera and laptop are not certain unless tested with. A Waterfall method that starts with a full-plan first, and revealing the shortfall of the model later in the project would be a huge risk. Therefore, Agile Method that works around the risky parts first is a safer choice since any problems will be detected earlier and there will be enough time to adjust the project development.

In this project, student takes the role of the product owner; developer and planner, and the supervisor acts as a stakeholder that provides reviews for each increment and gives feedback. As for the scrum master, the role would be assigned to the programme leader that checks on the progress of the student and assigned supervisor.

5.2 System Architecture

I. The Capture Layer

- Acquires live frames and Audio from the CCTV Camera/IP Camera over RTSP. Reading of frames with OpenCV and FFmpeg. The audio is resampled into 16 kHz mono and split into overlapping one second windows with 0.5 second hop.

II. The Detection Layer

- Runs pose estimation using YOLOv8-pose.
- Runs a cry detector using YAMNet embeddings with a small classifier head.
- Runs an anonymous tracker using Byte Tracker.
- Human-in-the-loop alert verification in case of false positives.

III. The Escalation Layer

- The detections are group into crying episodes and there will be an engine that will alert the user that is in charge.
- The user in charge will respond through a dashboard and acknowledge the situation, attend to the child or report as false alarm if the child is not crying. The choices made are counted as responses.

Parameter	Default	Meaning
Gap tolerance	30 seconds	Pauses in crying shorter than this keep the episode running. A longer silence than the gap tolerance given will end the crying episode and cancels potential alerts.
Alert threshold	5 minutes	Continuous crying within the gap tolerance exceeding alert threshold will trigger the first alert to the principal.
Response window	2 minutes	Time the principal has to respond before the alert is recorded as unanswered. If there's no response by the window, another alert will be sent in.

Owner threshold	More than 3 unanswered alerts	On the fourth unanswered alert within the same episode, the owner is notified with a signed escalation report on Telegram.
Recipients	Principal, then owner	Taken from a duty roster, so a named deputy can receive alerts when the principal is absent.

Table 2: System Parameter Table

IV. Integrity Layer

- Each stored clip is hashed into an append-only **SHA-256 chain.** This prevents deletion, insertion, or reordering of historical records without breaking the chain.
- **AES-256 (GCM)** encrypts footages. It ensures minors identities and CCTV records cannot be viewed if physical drives or cloud is leaked or broken into.
- The device uses **Ed25519 Signatures** to sign the block using its unique hardware private key. Anyone with the public key can verify the footage originated from that specific sensor.

5.3 Dataset and Ethical Considerations

For the violence recognition it will be benchmarked on available public datasets like the AudioSet cry-related classes, **ESC-50** and the **Donate-a-Cry corpus**. Hard negatives are drawn from the same sources as stated and publicly available audios for the background ambience of a nursery. Since the deployment context are nurseries, the individuals recorded would be minors. But, in this project, no real minors nor violence is used. Any original footages within the project are choreographed and performed by consenting adult volunteers that are above the age of 18. Otherwise, if there is an adult consenting their child for the testing, it would be filmed with compliance of ethics and per stated compliances in the consent form. A written consent will be provided and signed. Privacy safeguards like blurring are out of scope, so the prototype only remains on the local machine and not uploaded to public.

5.4 Threat Model for the Integrity Layer

Actor	Motive	Capability	Addressed by
Operator or staff member with file access	Remove or alter clip or record	Edit or delete files and database rows	Hash chain, signatures, append-only triggers, verifier.
Dishonest complainant	Present an altered clip as genuine	Modify an exported clip	Clip digests and signed log entries.
Outside attacker with system access	Cover tracks	Rewrite database and files	Separate key storage, external head-hash anchors.
Accident	-	Disk failure, power loss, camera outage	Integrity reports and backups.
Principal or deputy facing possible disciplinary action	Hide missed alerts or claim an alert never arrived	Delete entries, respond falsely, disable notifications or the camera	Delivery receipts, heartbeats, hash chain, individual credentials, and owner review of the raw evidence.
Owner or complainant acting unfairly	Use the record selectively against the principal	Present part of the log or a single clip	The verifier covers the full chain, the report lists every alert and response in the episode, and the principal can respond to the evidence.

Table 3: Threat Model Table

5.5 Privacy, Ethics and Regulatory Considerations

I. Children Data

The real deployment of system will capture infants, who cannot consent. In this prototype, there are no minors involved in the recording unless there is a parental permission for testing purposes which includes a permission form. Whereas for any real deployments, it would need an operator policy, parental notice and legal review.

II. Staff Monitoring

The system contains audio and visual analysis which will affect caretakers at their workplace. Any escalations will directly record and indicate whether named roles react to their responsibilities. The staffs should be informed in advance of what is being recorded, how the alert function and how the escalation records are used.

III. Recording Sensitivity

Given the system with crying detection, it processes live audio that may record ongoing conversations in the background. Staffs are informed beforehand on the audio recording of the system integrated in the CCTV so that they are well-aware of their conversations and way of behaving.

IV. Human Oversight

The system does not take any disciplinary actions. Its role is just to notify, escalate and record. Any further consequences for the staffs are based on human decision made by the owner of nursery and officials after the review of signed evidence and hearing the defence of person concerned.

6.0 UML Diagrams

I. ER Diagram

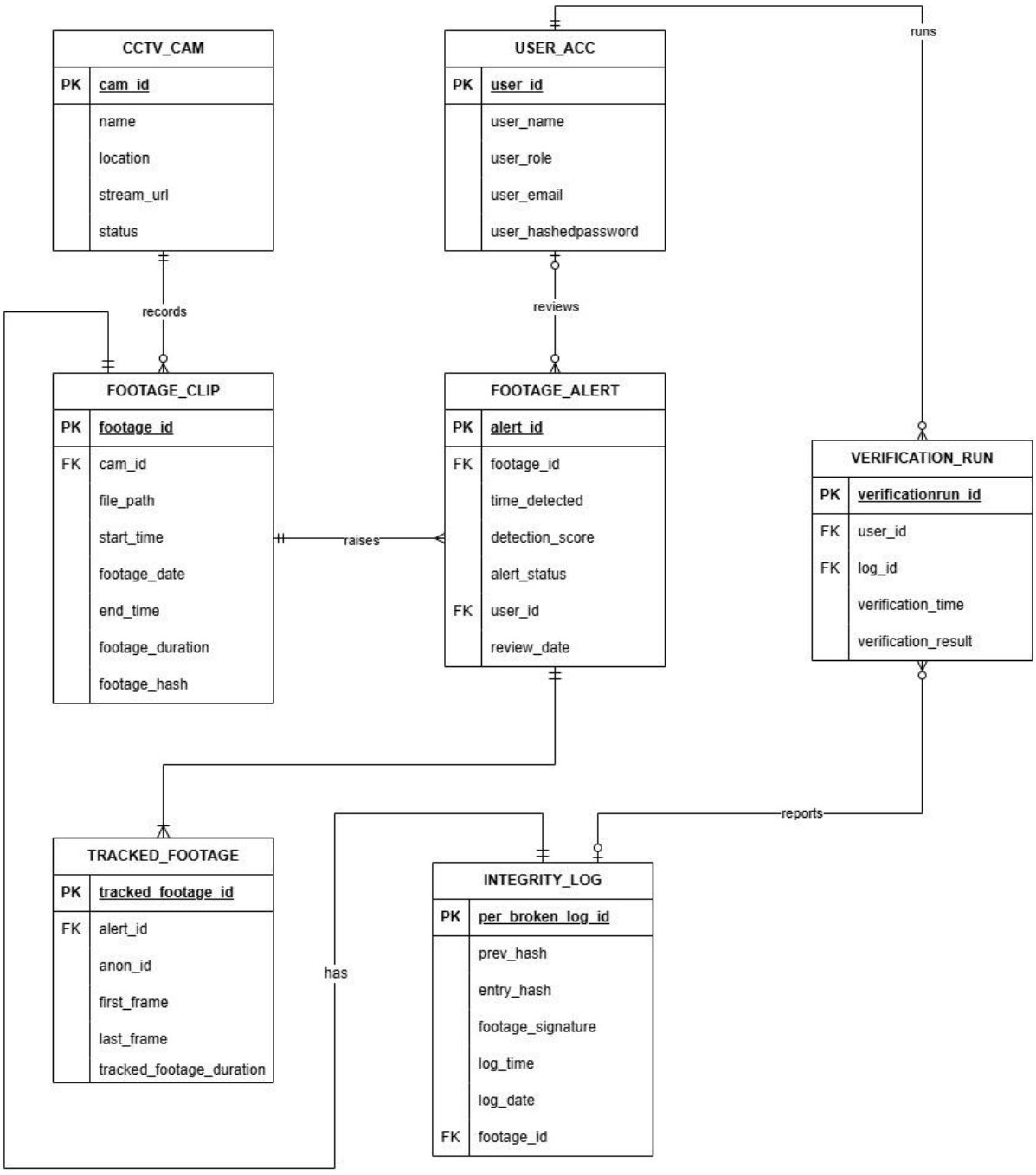


Figure 3: AI-Integrated CCTV System for Hazard Detection with Tamper-Evident Logging ER Diagram

II. Flowchart

i. Detection Flowchart

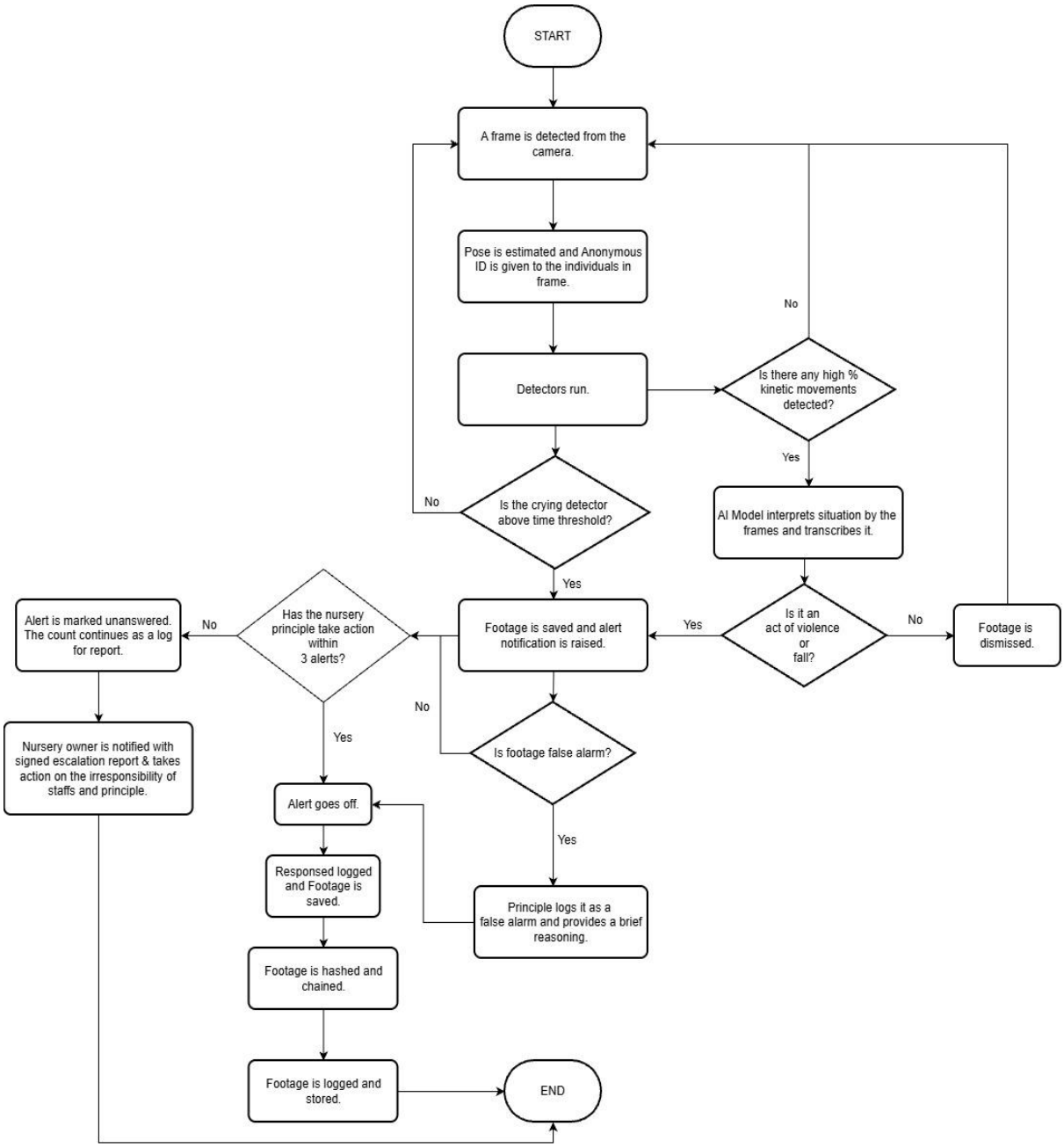


Figure 4: Footage Detection Flowchart

ii. Footage Verification Flowchart

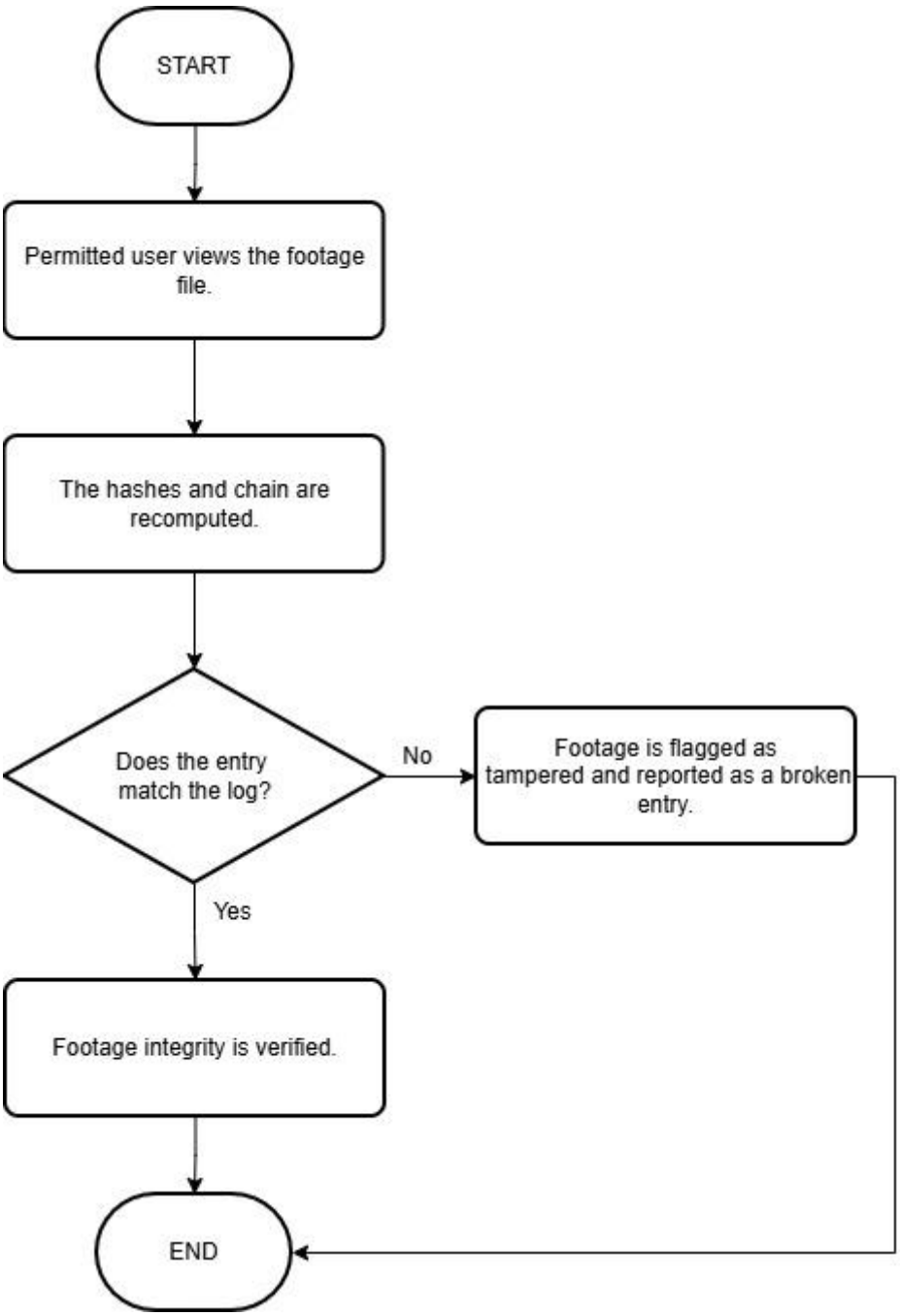


Figure 5: Footage Integrity Flowchart

III. Use Case Diagram

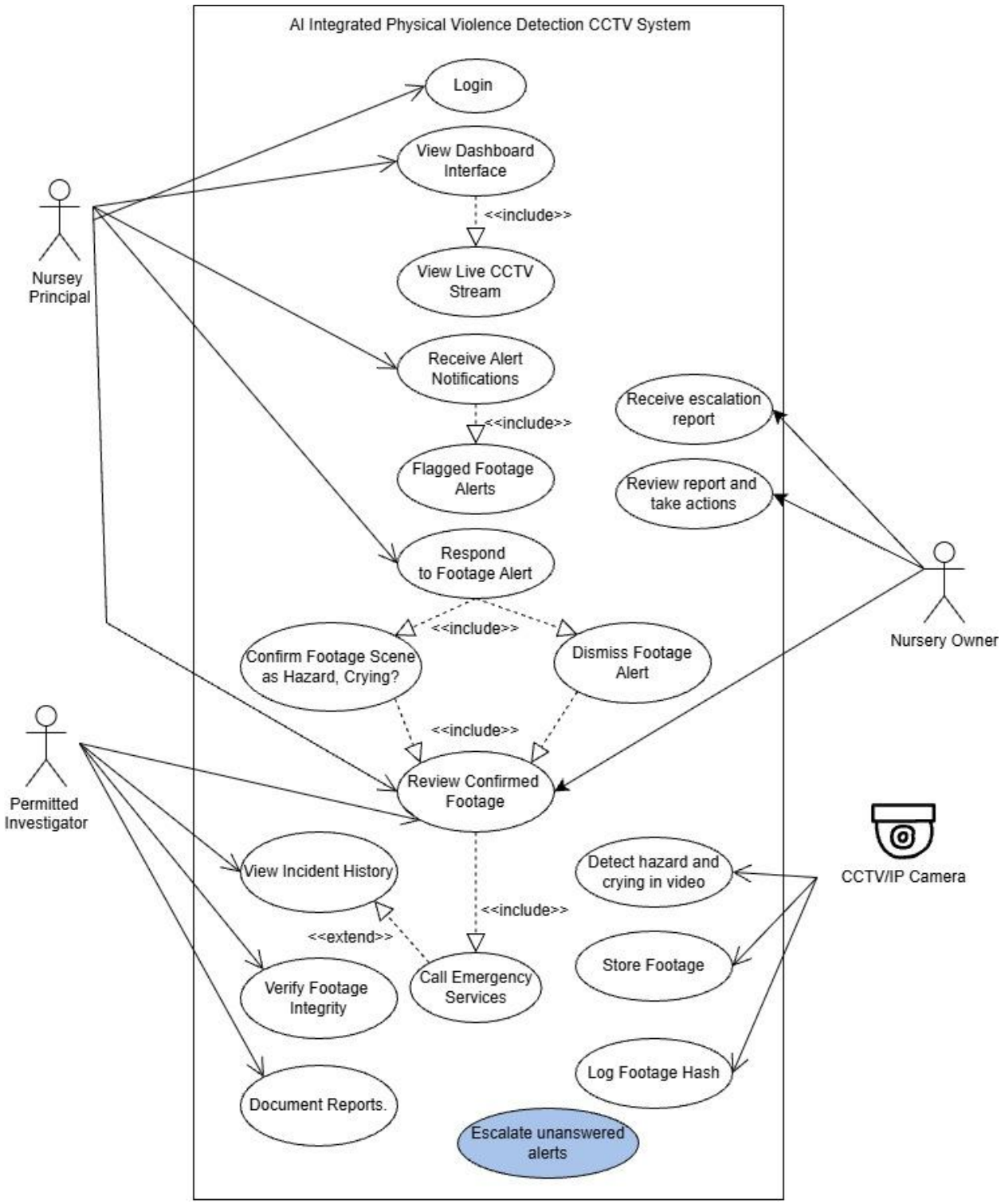


Figure 6: AI-Integrated CCTV System for Hazard Detection with Tamper-Evident Logging Use Case Diagram

7.0 Expected Deliverables

I. Hazard & Crying Detection Prototype

- A functioning prototype that detects hazardous events and infant crying from live camera audio. The duration of the crying is tracked and the principal is notified when the duration hits 5 minutes onwards. If there are no actions taken after 3 alerts with a 2-minute response window, the nursery owner will receive a notification on Telegram application regarding the issue.

II. Tamper-evident Footage Log:

- A signed, hash-chain log of events, alerts, responses and footage with a verification utility.
- A signed escalation report for the owner.

III. Source Code

- The source code for the system can be found in a version-controlled GitHub Repository.

IV. Documentation

- A project documentation containing the system design, architecture diagrams and user manual.

V. Presentation Poster

- During the final viva, a poster must be presented at the respective booths for a poster presentation and exhibition of the project.

VI. Final Presentation & Report

- During the final viva, a summarised project report, poster and presentation slides will be prepared for evaluation of project.

8.0 Risk Planning

Risk	Likeli-hood	Impact	Mitigation
Public datasets are small and do not match nursery acoustics.	High	High	• Combine several sources, use pretrained backbones and build a mock-nursery test set.
Detector confuses toddler shouting, laughter or other sounds with crying.	High	High	• Hard-negative training, human verification, and honest reporting of false alarms. • Implement LLM into the system for detecting the sounds.
Camera does not expose audio over RTSP, or audio quality is poor.	Medium	High	➢ Revert to a smartphone IP-camera app or a separate USB microphone.
Clock drift weakens timestamp evidence.	Medium	Medium	➢ NTP synchronisation and logging the time source.
Escalation counts alerts unfairly (false alarm, principal absent, message not delivered).	Medium	High	➢ Every alert links to a reviewable clip, false-alarm responses count as responses, a duty roster names a deputy, delivery receipts are logged, and the owner reviews evidence before any action.
Notification channel fails or is delayed.	Medium	High	➢ Use the dashboard with an addition of a second channel, log delivery receipts, monitor the notification service with heartbeats, and show failures in the escalation report so they are not attributed to the recipient.

Table 4: Risk Planning Table

Risk Assessment 3x3 Matrix:

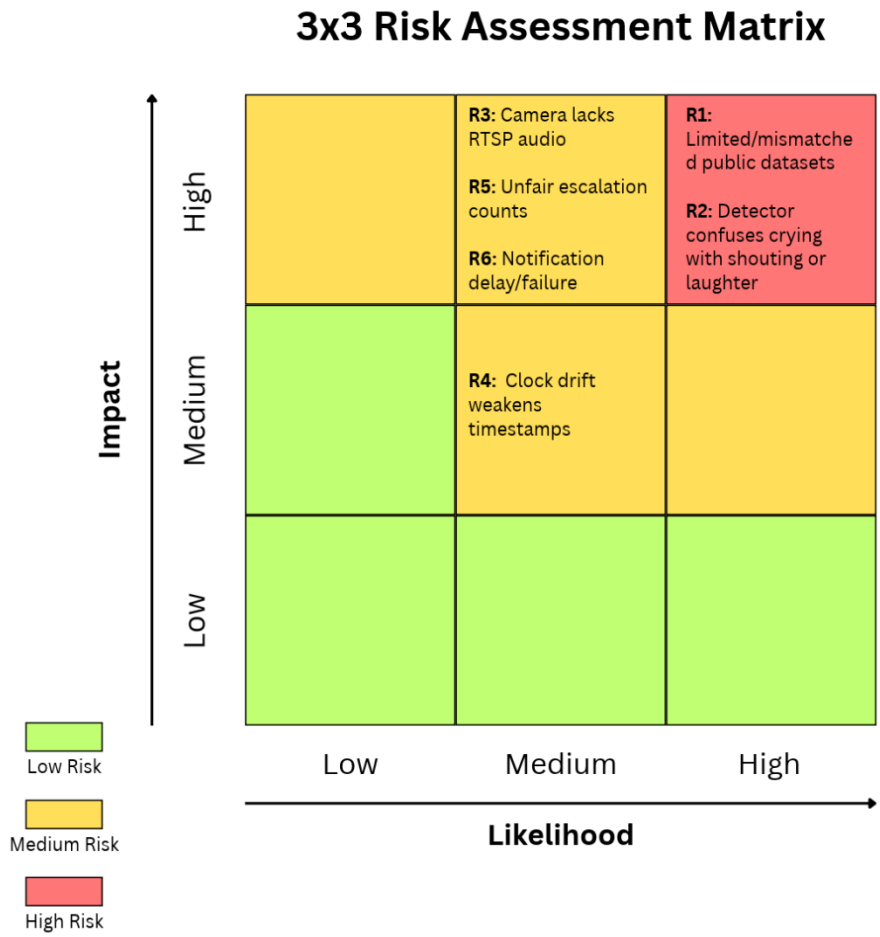
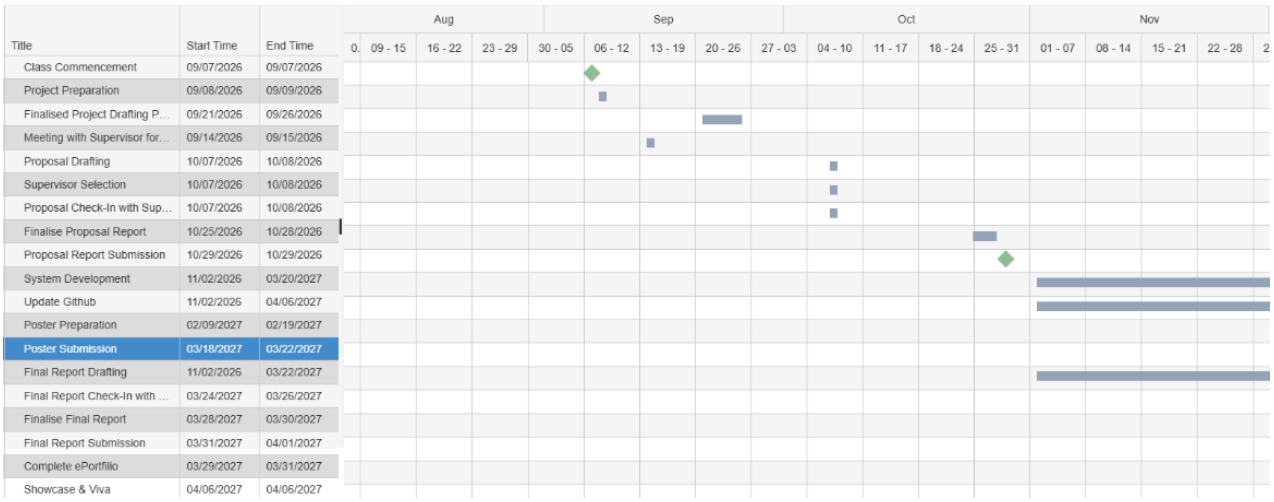


Figure 7: 3x3 Risk Assessment Matrix

9.0 Proposed Gantt Chart



Title	Start Time	End Time
Class Commencement	09/07/2026	09/07/2026
Project Preparation	09/08/2026	09/09/2026
Finalised Project Drafting P...	09/21/2026	09/26/2026
Meeting with Supervisor for...	09/14/2026	09/15/2026
Proposal Drafting	10/07/2026	10/08/2026
Supervisor Selection	10/07/2026	10/08/2026
Proposal Check-In with Sup...	10/07/2026	10/08/2026
Finalise Proposal Report	10/25/2026	10/28/2026
Proposal Report Submission	10/29/2026	10/29/2026
System Development	11/02/2026	03/20/2027
Update Github	11/02/2026	04/06/2027
Poster Preparation	02/09/2027	02/19/2027
Poster Submission	03/18/2027	03/22/2027
Final Report Drafting	11/02/2026	03/22/2027
Final Report Check-In with ...	03/24/2027	03/26/2027
Finalise Final Report	03/28/2027	03/30/2027
Final Report Submission	03/31/2027	04/01/2027
Complete ePortfilio	03/29/2027	03/31/2027
Showcase & Viva	04/06/2027	04/06/2027

Figure 8: Project Gantt Chart (1)

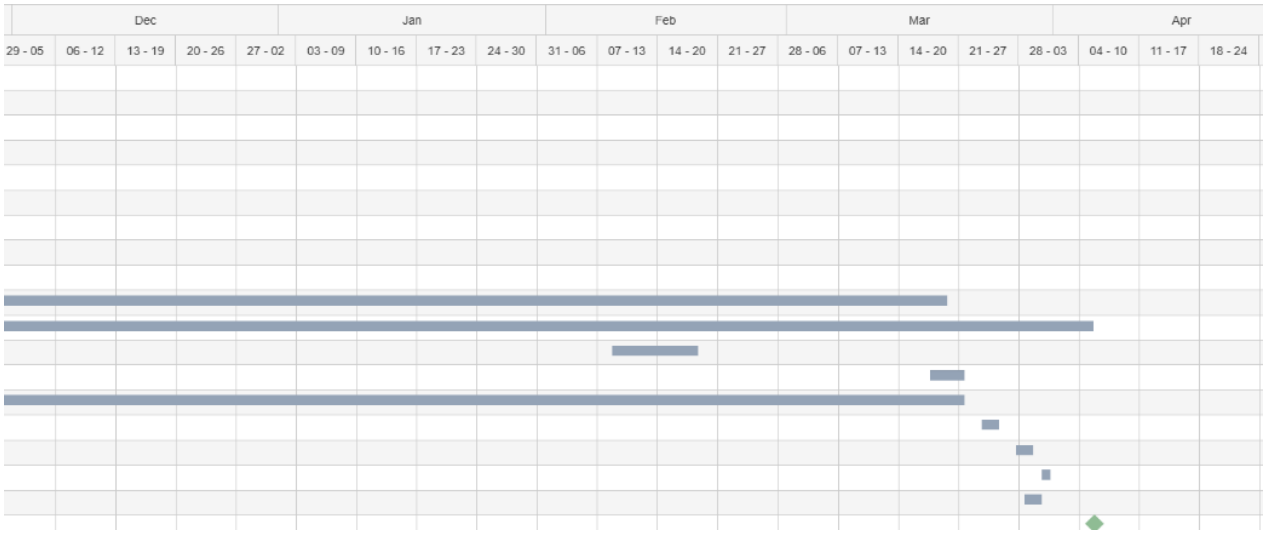


Figure 9: Project Gantt Chart (2)

10.0 Resources Requirements

➢ Hardware Requirements

Table 5 shows the hardware requirements that are important for the project development.

Category	Tool	System Name	Purpose
Local machine (Laptop)	64-bit operating system, x64-based processor	Intel(R) Core (TM) i3-10110U CPU @ 2.10GHz (2.59 GHz)	For coding, development testing and documentation.
Camera (Development)	Smartphone with an IP camera application.	-	Used for the early-stage development. Also works as a backup camera in case of CCTV not functioning.
Close Circuit Television (CCTV) Camera (Main Utility)	TP-Link Tapo C210 (3MP, 2304 x 1296, H.264, RTSP/ONVIF, 2.4GHz Wi-Fi)	-	Used as a medium of live footage capturing.
Local machine Webcam	-	-	Used for Stage 1 and Stage 2 development. An alternative to the CCTV and IP Camera in any cases of non-functionality.
Speaker	Existing speakers.	-	Playback of licensed cry and noise clips in staged scenes.

Table 5: Hardware Requirement Table

- **Software Requirements**

Table 6 shows the software requirements that are crucial for the project development.

System Name	Purpose
YOLOv8-pose	Used for Pose Estimation.
ByteTrack	Used for Anonymous Tracking.
OpenCV	Used for Camera Input (RTSP).
SQLite, hashlib and cryptography library	Used for Hash-chain logging and verifier.
Streamlit/Flask	Used to create the dashboard and alert verification.
Python	Main programming language used for the development.
PyCharm Community Edition	IDE used for writing the source code.
GitHub	Used to upload incremental progression on source code and documents.
Figma	Prototyping the interface and dashboard for a visualised framework.
Kaggle	Medium for downloading publicly available datasets.
Telegram Bot API	Alert delivery to the principal and owner, with response links and delivery receipts.

Table 6: Software Requirements Table

- **Other Requirements**

Table 7 shows the libraries that are needed to be installed in the IDE for running the source code.

Library Name	Version
numpy	2.5.4
pandas	3.0.6
Torch	2.6.0+cu124
Qwen-vl-utils	0.0.14
Pillow	12.3.0
Pydantic	2.14.1
Transformers	5.20.2
Accelerate	1.15.0
librosa, torchaudio, FFmpeg	-

Table 7: Other Requirements Table

11.0 Expected Outcome and Significance

Once the system is fully implemented, the AI Integrate CCTV system aims to being impactful and brings a meaning to raise confidence of nursery operations and non-neglection towards the children within the premise. Its primary goal is to create a system that will detect any hazardous events and detect crying that exceeds 5 minutes with a live alerting system that will alert the nursery principal. With a reactive system, the staffs and principal would not be aware of every single child in the premise as there is a number ratio of number of children to a caretaker. On the other hand, with the tamper-evident cryptographic system to verify whether the footages have been tampered with or not, plays a role in proving that there is integrity in the footage data.

The system developed is also significant as it benefits the children and parents at the same time. As for the children, their life is at less risk as the system is actively monitoring the activities of the children. Fast alerts by the system and quick action taken by the person-in-charge (PIC) have huge contribution to ensuring the well-being safety of each and every child. Besides, parents would feel less heavy hearted to send their child to nurseries due to fear of their child's well-being and sanity. They will be at ease knowing that their children are in a safer environment while being actively monitored.

Therefore, a reactive system like the **AI-Integrated CCTV for Hazard Detection with Tamper-Evident Logging in Nursery and Child Care System** would play a huge role for Malaysia's nursery safety.

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